On 'Astronomy and Space Science'

for HKDSE

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Abstract

Before the developement of telescopes and the modern astronomical technologies, the ancient Greek postulated "the Earth is the center of the cosmos" as one of their axioms. This, of course, we now know is untrue. Yet around 16-18th centuries, many predictions were made by polymaths at the time, such as Galileo Galilei (1564-1642), Leonhard Euler (1707-1783), Johannes Kepler (1571-1630). Many of which are still used in today's understanding of astronomical bodies (and beyond) and has been tested to great accuracy, despite many being originally simple emperical observation by looking at the night sky. In this lecture series, we will discovery and explore these remarkable ideas.

In order to relate directly to the HKDSE course, we will use their curriculum's topic titles as section titles in this lecture note. Learning outcome expected will also be listed in the beginning of each chapter. These notes will be continually updated as the lecture series progresses. Please do not hesistate to contact me should you spot any mistakes/have any remarks.

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Chapter 1

The Universe As Seen In Different Scales

1.1 Learning Outcomes

- use the "Powers of Ten" approach to describe the basic features and hierarchy of celestial bodies such as satellite, planet, star, star cluster, nebula, galaxy and cluster of galaxies, as seen in different spatial scales
- define the basic terminologies such as light year and astronomical unit for describing the spatial scale

1.2 Units and Orders of Magnitudes

Astronomy is a physical science. To bring mathematics to physical world, we must somehow relate them via some fixing of values. For example, in Electrodynamics we introduce ε_0 and μ_0 (i.e. permittivity or permeability of free space); or in Quantum Physics, we introduce $\hbar$ (Planck's constant); or in Newton's Gravity, we introduce G (Newton's Gravitational constant) etc. It then comes to the question: "well what do we introduce for astronomy?".

Apart from its obvious connotation with Astronomy, the word "Astronomical" can also mean things of massive, extreme magnitude. Think: the Sun is 333,000 times greater in mass than Earth; Milkyway is 500,000,000,000,000,000 times greater than Earth. This is *astronomical*. Hence, Astronomer adopts units that "normalise" to this by picking a reference that is equally *astronomical*.

1.2.1 Mass

Definition 1.1 (Solar Mass)**.** *In Astronomy, it is conventional to define mass in terms of Sun's. This is known as "Solar Mass", denoted by $M_\odot$, where it converts to SI unit (kg) by:*

$$M_\odot \equiv 1.989 \times 10^{30}\ kg$$

Remark 1.1. *We say mass of the Sun is unit mass in Solar Mass system, that is mass of the Sun in solar mass is $1M_\oplus$.*

From which it follows the equivalances as examples:

$$M_\odot = 3.33 \times 10^5\ M_\oplus$$

$$M_\odot = 1047.35\ M_J$$

where $M_\oplus$ and M_J denotes mass of the Earth and Jupiter respectively.

Question 1: Conversion Between Mass Systems

a) Express mass of Jupiter M_J in terms of $M_\oplus$

b) Express mass of the Milky Way in terms of $M_\odot$

c) Express $m = 3.978 \times 10^5$ kg in terms of $M_\oplus$

1.2.2 Distance

Since in Astronomy, distance considered are again often large in magnitude. It is conventional to select lightyear as a unit of distance. Furthermore, this has origin from theory of Special Relativity which, although highly relevant, is out of the scope of this course and hence omitted from the discussion in length.

Definition 1.2 (Lightyear)**.** *The distance travelled by light (in a vacuum) in a year is defined as "lightyear". In other words:*

$$\text{ly} = \underbrace{3 \times 10^8 \ m \ s^{-1}}_{c} \underbrace{60 \times 60 \times 24 \times 365.25}_{year \ in \ second} = 9.46 \times 10^{15} \ m$$

Equally as common, again, we can define distance in terms of the Sun:

Definition 1.3 (Astronomical Unit)**.** *The distance between the Sun and the Earth is the astronomical unit, denoted au, that is:*

$$1 \ au \equiv 1.50 \times 10^{11} \ m = 1.50 \times 10^8 \ km$$

Question 2: Conversion Between Distance Systems

Express the following in terms of i) metre (m), ii) lightyear (ly), iii) astronomical unit (au), and iv) comment on the choice of systems. Which one(s) is/are more sensible for what cases?

a) Radius of Earth

b) Radius of the Milky way

c) Bohr's radius

N.B.: Consult online sources for these distances.

1.3 Celestial Bodies

Here, we will discuss some structures for celestial bodies and their respective characteristics. A summary for their magnitude is given in Table (1.1).

Satellite

- **Definition 1.4** (Satellite)**.** *An object that orbits a larger celestial body (a planet or dwarf planet).*
- Satellite can be categories into 2 main types:
  - *Natural satellites:* Moons (e.g., Earth's Moon, Europa orbiting Jupiter).
  - *Artificial satellites:* Artificial objects placed into orbit for communication, weather monitoring, or scientific research (e.g. International Space Station).

Question 3: Period of Satellite

1. Derive an expression for the period of a satellite of mass m_{stat} at height h above Earth's surface with mass $M_\oplus$. (Hint: the distance is **NOT** simply h.)
2. Hence or otherwise, express the above in terms of solar mass $M_\odot$ and astronomical unit (au).

Planet

- **Definition 1.5** (Planet)**.** *A celestial body that:*
  1. *Orbits a star (or stellar remnant).*
  2. *Has sufficient mass to assume a nearly round shape (hydrostatic equilibrium).*
  3. *Has "cleared the neighborhood" around its orbit, i.e. large enough to clear away other objects of similar size from its orbital path*
- Plants also have types:
  - *Terrestrial planets:* primarily composed of rock, silicate, water and/or carbon → rocky planets!
  - *Gas Giant (/Jovian) planets:* primarily composed of low-boiling point materials, rather than rock or other solid matter → gassy planets!

Star

- **Definition 1.6** (Star)**.** *A massive, luminous spheroid of plasma held together by self-gravity.*
- Often, some objects orbit around a star:
  - *Asteroids:* small rocky objects, mostly located in an asteroid belt between orbits of planets
  - *Comets:* small icy objects, displays a long luminous tail that points away from the star when it is close to it.

Star Cluster

- **Definition 1.7** (Star Cluster)**.** *Large groups of stars bound together by mutual gravitational attraction.*
- **Main Types:**
  - *Open Clusters:* Loose groups containing up to a few thousand young stars, typically found in the disk of a galaxy (e.g., The Pleiades).
  - *Globular Clusters:* Tightly bound, spherical collections of tens of thousands to millions of older stars, usually found in the galactic halo.

Nebula

- **Definition 1.8** (Nebula)**.** *An interstellar cloud of dust, hydrogen, helium, and other ionized gases.*
- They are basic bulding block of the universe:
  - Mostly hydrogen (75-90%) and helium (think organic chemistry!)
  - Small amount of other elements which comes from stellar nucleosyntheis.

Size (in power of 10)	Celestial Object
10^6 m	Satellite (e.g. Moon)
10^7 m	Terrestrial Planet
10^8 m	Gas Giant Planet
10^9 m	Star
$10^{15} \sim 10^{17}$ m	Nebula
$10^{17} \sim 10^{18}$ m	Star Cluster
10^{21} m	Galaxy
$10^{23} \sim 10^{24}$ m	Galaxy Cluster

Table 1.1: Celestial Hierarchy Ordered by Spatial Scale

Galaxy

- **Definition 1.9** (Galaxy)**.** *A massive, gravitationally bound system consisting of stars, stellar remnants, interstellar gas, dust, and dark matter.*
- It contains stages of star:
  - Neblae
  - Star clusters
  - Main sequence stars, and their plantary system
  - Stages of star: red gaint, white dwarfs, neutron stars, blackholes etc.[1]

Cluster of Galaxies

- **Definition 1.10** (Galaxy Clusters)**.** *A structure consisting of hundreds to thousands of individual galaxies bound together by gravity.*
- **Superclusters:** Clusters of galaxies can further group into larger structures called superclusters, separated by vast cosmic voids.

Question 4: Celestrial Bodies

1. For each category, give an example of them. For example, Earth is a planet and the Sun is a star.
2. State their respective size in power of 10.
3. Form a ranking from largest to the smallest for these examples.

Question 5: Solar System

We live in the solar system, let's look at it in more detail:

1. Name the planets within the Solar System and arrange by distance from the Sun (Hint: "*My Very Easy Method Just Speeds Up Naming*")
2. Further add details by group the type of planets. (Hint: which ones are rocky (terrestrial)? which ones are gassy (Gas Giant)?)
3. Pluto used to be part of the Solar System. However, in 2006, Pluto was removed in the categorisation... Research why

[1] More discussion to follow during chapter on H-R Diagrams.

Chapter 2

Astronomy Through History

2.1 Learning Outcomes

- compare the heliocentric model with the geocentric model in explaining the motion of planets on the celestial sphere
- describe Galileo's astronomical discoveries and discuss their implications
- describe planetary motion using Kepler's laws

2.2 Heliocentric, Geocentric, and the Renaissance

2.2.1 Nicolaus Copernicus (1473-1543)

Nicolaus Copernicus proposed that the Sun is the center of the planetary system:

- Known as the **Heliocentric model** of the universe.
  - "helio-": from Greek, *helios*, meaning "Sun".
  - "-centric": from Greek, *kentrikos* (pertaining to a center), derived from *kentron* ("sharp point" or "center").
- Designed to explain planetary **retrograde motion** (as an optical illusion when Earth overtakes outer planets) and varying planetary brightness without relying on complex epicycles.
- The dominant authority at the time, supported by the Catholic Church, adopted the **Geocentric model** (Ptolemaic system), which placed Earth at the center.
- His work was placed on the Church's *Index of Forbidden Books* much later, in 1616.

2.2.2 Galileo Galilei (1564-1642)

Galileo Galilei (yes that's his actual name!) provided the crucial observational evidence supporting the Heliocentric model:

- Used an improved optical instrument called a **telescope**.
- Discovered four moons orbiting Jupiter (the Galilean moons), proving that not all celestial bodies orbit the Earth and that other centers of motion exist.
- Observed the complete cycle of **phases of Venus** (accompanied by changes in apparent size), providing direct observational proof against the strict Geocentric model.
- Discovered craters/mountains on the Moon and sunspots on the Sun, disproving the ancient Aristotelian dogma of perfect, unchanging celestial bodies.
- The Church hated this of course: Formally tried by the Roman Inquisition in 1633, forced to recant his support for heliocentrism, and spent the remainder of his life under house arrest.

2.3 Kepler's Three Laws

The German polymath and astronomer Johannes Kepler (1571-1630) formulated three laws describing planetary motion using Tycho Brahe's (Danish astronomer) observational data:

- **First Law (Law of Orbits):** The orbit of a planet is an ellipse with the Sun at the foci.
- **Second Law (Law of Areas):** Planets on the orbit sweeps out exactly the same area in equal time interval. That implies planets move faster at **perihelion** (closest to the Sun) and slower at **aphelion** (farthest from the Sun).
- **Third Law (Law of Periods):** The square of the orbital period (T) of a planet is directly proportional to the cube of the semi-major axis (R) of its orbit, expressed as $T^2 \propto R^3$ (or $\frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3}$).

The remarkable thing about these results is that: they were all obtained initially through simply emperical observation. That is, tis is not done through any theoretical prediction. Furthermore it successfully bridged the gap between Copernicus's heliocentric concept and rigorous mathematical physics for Newton's law of gravitation.

Question 6: Kepler's Law

To help understand Kepler's Law more intuitively:

1. State each law and draw a/some diagrams them to demonstrate their idea.
2. Show that Kepler's 3rd law is true from Newton's Law of Gravitation.
3. What is an ellipse? What are some associated properties of it?
4. Although many technologies have been upgraded from Kepler's time, many of the nature of scientific collaboration remains similar. Kepler used others dataset to conclude his laws. Can you name examples of modern day Physics collaboration and their contribution?

Chapter 3

Orbital Motions Under Gravity

3.1 Learning Outcomes

- Newton's law of gravitation
  - apply Newton's law of gravitation $F = \frac{GMm}{r^2}$ to explain the motion of celestial bodies in circular orbits
  - derive Kepler's third law $T^2 \propto r^3$ for circular orbits from Newton's law of gravitation
  - state Kepler's third law for elliptical orbits $T^2 = \frac{4\pi^2 a^3}{Gm}$
  - apply Kepler's third law to solve problems involving circular and elliptical orbits
- Weightlessness
  - explain apparent weightlessness in an orbiting spacecraft as a result of acceleration due to gravity being independent of mass
- Conservation of Energy
  - interpret the meaning of gravitational potential energy and its expression $U = -\frac{GMm}{r}$
  - apply conservation of mechanical energy to solve problems involving the motion of celestial bodies or spacecraft
  - determine the escape velocity on a celestial body

3.2 Newton's Law of Gravitation

Theorem 1 (Newton's Law of Gravitation)**.** *The mutual gravitational attraction, i.e. gravitational force, between two separated masses of mass M and m for distance r respectively is given (in magnitude) by:*

$$F = \frac{GMm}{r^2} \tag{3.1}$$

Remark 3.1. *Since F in general is a vector quantity, attention to the sign must be taken when we are not dealing with simply its magnitude.*

Consequently, the Kepler's law is straightforwardly dervied from this:

Proposition 1 (Kepler's 3rd Law)**.** *For a circular orbit with radius r, its period follows the proportionality relation:*

$$T^2 \propto r^3 \tag{3.2}$$

Proof. By relating Newton's law of gravitation and the centripetal force:

$$F = \frac{GMm}{r^2} = m\omega^2 r \tag{3.3}$$

Note that mass dependence can be dropped:

$$\frac{GM}{r^3} = \omega^2 = \left(\frac{2\pi}{T}\right)^2 \tag{3.4}$$

Rearrange to get:

$$T^2 = \frac{4\pi^2 r^3}{GM} \propto r^3 \tag{3.5}$$

□

Question 7: Kepler's Law (again!)

Extend the previous result to elliptical orbit

1. Draw the diagram to show what a represents in $T^2 = \frac{4\pi^2 a^3}{GM}$
2. Comment on why m has no dependence here: we know that it cancels out from the Maths but why should this be true physically?

3.3 Weightness

The answer to Question 7.2 is the core proposition of this section. Imagine the following scenerio:

Question 8: Falling in a lift

Suppose you are person of mass m, standinh inside a lift of mass M. Initially, the lift is suspended by a light inextensible cable and is at rest at time $t = 0$. Take the downward acceleration due to gravity to be g.
With this scenario, answer the following structured questions:

1. (a) State the Newton's 2nd Law and why we need it here. Is the law determinstic? (Hint: determinstic means the outcome is completely predictable and fixed by initial conditions)
   (b) Draw a labeled free-body diagram showing all the forces acting on the person at $t = 0$.
   (c) Hence, write down the magnitude of the normal reaction force R exerted by the lift floor on the person.
2. At a later instant, the cable snaps and the lift falls freely downwards.
   (a) State the acceleration of the person and the lift during free-fall.
   (b) By applying Newton's 2nd Law to the person, derive an expression for the new normal reaction force $\tilde{R}$ acting on the person.
3. Define a frame of reference. Hence or otherwise, Based on your result in part 2(b), describe and explain the sensation experienced by the person regarding their weight. What can you conclude by comparing your experience in the lift's frame and your own?
4. Explain whether the person's actual (gravitational) mass has changed during the free-fall motion. Is it frame dependent?
5. Following from the result of part 4), can you name examples of quantity that are frame-dependent (i.e. value changes dependent on frame) and frame-independent?

The above is used by Albert Einstein (1879-1955), or many similar analogues (e.g. a spaceship instead of a lift), to ultimately derive his Theory of Relativity. The remarkable thing here is that, despite a very simple assumption that "physical law is same in all frame of reference", the consequence of General Relativity allows us to predict many important results of Cosmology (e.g. expansion of the universe), technologies (e.g. GPS) and even in films: the movie 'Interstellar' (2014), directed by Christopher Nolan, has famously a scene of a computer simulated animation of black hole (see here).

To read more about this, visit this page on (Weak) Equivalence Principle(s).

3.4 Conservation of Energy

To follow in the next lesson.